

Antiderivatives

section 4.2

10/17/25

Key Word:

INTEGRAL

lemme say it again

INTEGRAL

"definite integral"



find area from a to b under $f(x)$

how to:

- Split into rectangles

Key Notation

$$\int_a^b f(x) dx$$

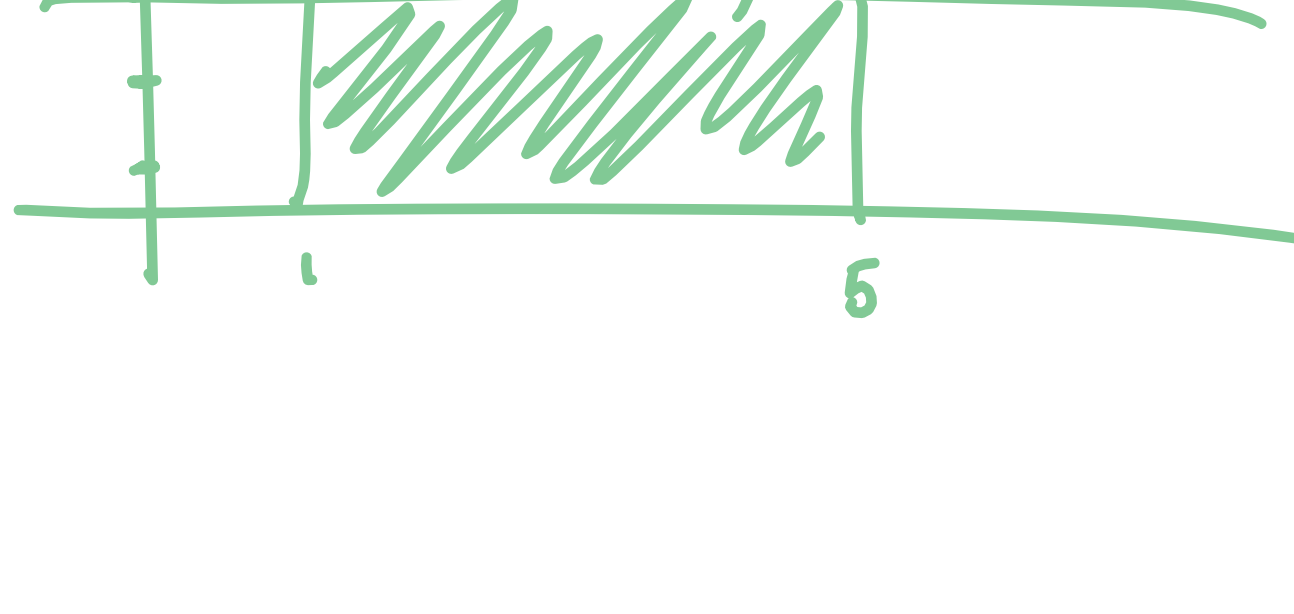
"Indefinite Integral"

$$\int f(x) dx$$

NO A or B, NO range

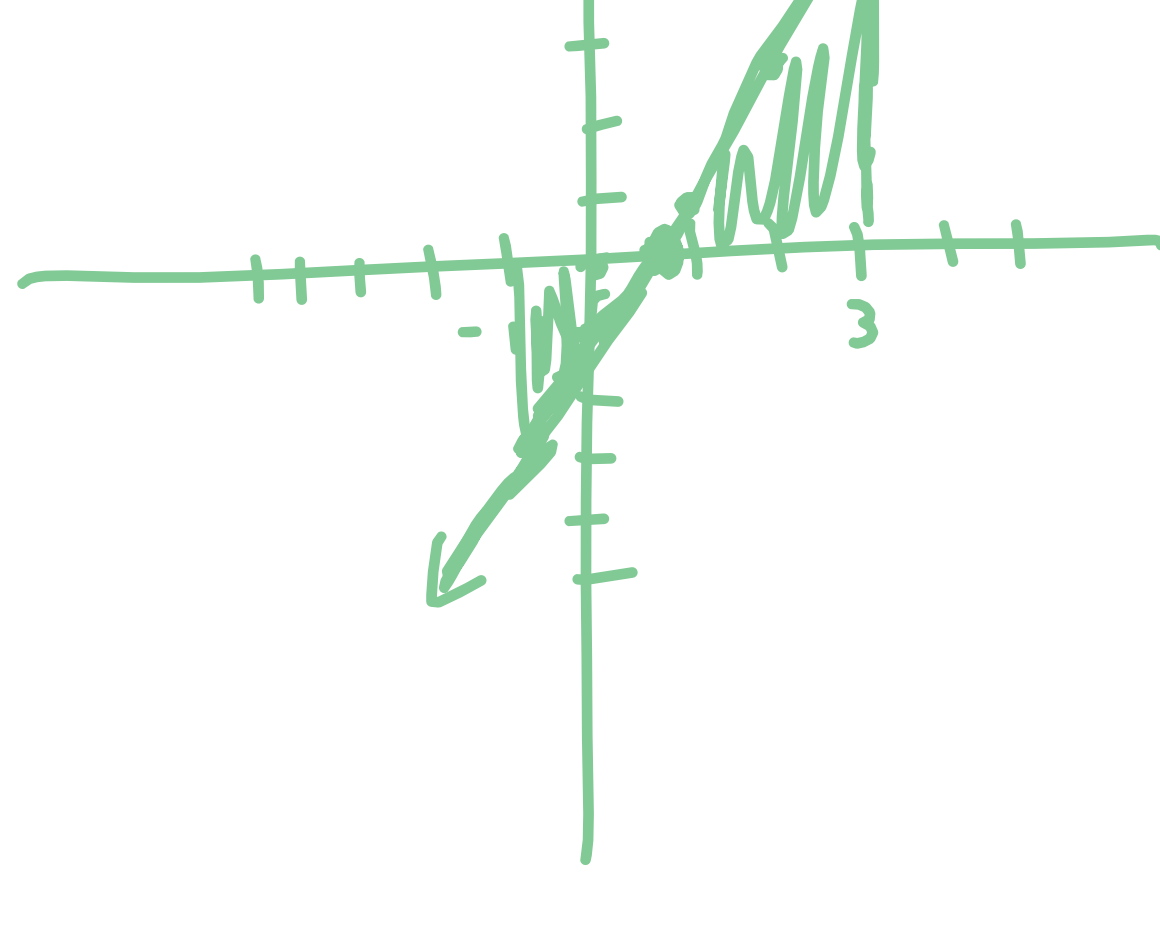
Ex 1:

$$\int_1^5 3 dx = 12$$



Ex 2:

$$\int_{-1}^3 2x-1 dx = 4$$



$$\frac{1}{2} \left(\frac{9}{2} \right) (-3) + \frac{1}{2} \left(\frac{5}{2} \right) 5$$

$$-\frac{9}{4} + \frac{25}{4} = \frac{16}{4} = 4$$

Integral Properties

$$1) \int_a^a f(x) dx = 0$$

$$2) \int_b^a f(x) dx = -\int_a^b f(x) dx$$

$$3) \int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

$$4) \int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$5) \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

dist. even

$$6) \int_{-a}^a f(x) dx = 0$$

dist. odd